Liverpool association for Space engineering & research...

LASER Newsletter

Mach-X team updates



Avionics system diagram

It's been a productive week at Mach-X, with plenty of progress being made through the week. One big milestone is that Avionics has completed their order forms! This officially means that they can begin the procurement process, which is fantastic news. At the moment the team is prioritising the main board as 'LIFTSv3', this is because the main board is essentially the brains of the rocket, and therefore will be instructing all of the other components. The Cansat Canbus schematic we talked about last week is also almost finished, as well as the launch vehicle team making key advancements on their rocketPy code. This improvement in the code will allow for more accurate simulation of their airbrake system, which is vital to accurately visualising the control the team has over its altitude mid flight. And lastly CAD for the rocket design is still in the works, but is looking in good shape and is something we look forward to spilling more on next week.

Riddle Of the week...

I am behind you, infront of you & all around-
What am I ?

last weeks answer: light

answer posted next week

WIC 9th November 2026

NRC team updates

Unity rise have a game plan! Earlier this week Filip and Aiden sat down to create a goal-oriented time line for the team to help organise resources from now to launch day. Putting the plan to work, the launch vehicle team have been experimenting with preliminary rocket design, and designing CAD models for these prototypes. The avionics team are continuing to research sensors, with a prime focus on temperature and humidity tracking, the idea is that this will help provide purpose to the rocket, allowing it to track weather. This brings the team closer to completing their first draft, which will be their goal for the next few weeks.



Jacob hard at work on CAD

Rover team updates

The rover team is moving along steadily, with applications being closed last week. Unfortunately, we cannot confirm names or roles this week, but hopefully we'll have more information to update you on next time! Just building the anticipation...
Congratulations to all who were accepted!



What's up in the space & engineering world?

Students v Moon dust, new landing pads testing to improve lunar landing saftey

Moon landings are well known for being unsafe and messy, but this week students, like us, have been making huge advancements to try and decrease the risk and tackle moon dust head on.

When rockets land vertically the heat from the engines and force from the rocket causes dust and rocks, colloquially known as moon dust, to be disturbed and cloud the landing site. Although moon dust sounds soft and whimsical, it is actually razor sharp, abrasive, and chemically reactive due to the lack of atmosphere to dull the shards. This debris essentially acts as a sandblaster on the rocket causing damage to hardware and equipment, often getting stuck in joints and clogging mechanics. It also obviously isn't great for visibility.

Now why I think this article is interesting is because people our age at Alabama University (supported by NASA) are working hard to improve moon landings and reduce moon dust. The team has constructed vented concrete components that come together to form an interlocking landing pad that also includes channels to mitigate heat. The concrete used in the prototype is a place holder to simulate known as 'lunacrete', which is concrete made using lunar regolith (lunar soil), with hopes that the landing pad can be constructed on the moon.

The landing pad prototypes have already gone under initial testing, using motor-firing at the universities Propulsion Research Center. This means that they can now work on larger models and builds to help determine if this method can be used on a larger scale, and challenging the final design.

This marks a major step in lunar infrastructure development, as well as marking a huge achievement for students in STEM. The hope of the students is that overtime, due to the materials being made from things available on the moon, that these landing pads will be easier to expand and repair and make moon landings more accessible.

Prototype testing



landing pad concept design



Weekly event updates

13th of November personal project meeting
*Mandatory for project participants
→risk assesment info
→ presentation writing skills
→order form protocol

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